



Chinese Military Gaming Bus

To avoid feeling isolated and bored, the Chinese military has its own camouflage bus meant only for gaming <http://dgit.in/MilitaryGaming>



Singing Tesla Coil

TinyTesla is a device anyone could use to build their own Tesla coil and play MIDI music files on it <http://dgit.in/TinyTeslaCoil>

Demigods of computing you may not have heard of

We look at ten individuals from the past whose inventions and concepts were crucial milestones, paving the way for computing in the 21st century

Abhijit Dey
readersletters@digit.in

We've all heard about Bill Gates, who created Microsoft along with Paul Allen and revolutionised the software industry; Steve Jobs, who founded Apple Computers with Steve Wozniak and commercialised the entire personal computing industry; and Tim-Berners Lee, who created the World Wide Web and completely transformed what the Internet could do. But there are many more pioneers who made seminal contributions to computer science that we don't know about. Trailblazers whose names got lost in the plethora of names associated with computing. Here are some of those demigods who have created some of the most important things in this domain.

1 Grace Hopper (Compiler, COBOL)



After a brief tenure of working with the US Navy, Grace Hopper created the first ever compiler for a computer programming language in 1951 when programmers used machine language for writing code. A compiler converts the source code written in a programming language into a machine-understandable code for the machine to execute. Initially, computers were programmed in binary machine code, and only a limited number of people could really understand such code and write it. The compiler encouraged further development not only for programming but also for the computer, which was then

used only as a calculator. Grace wanted programming languages to be closer to the English language, so that we could learn and understand them better. And hence in 1959, she went on to develop COBOL, which was one of the first modern programming languages that still finds application in certain fields. The term 'debugging' was popularised by her when her team found an actual bug (a moth, to be specific) in the wiring of their computer system.

2 Ajay Bhatt (USB)



"Our rockstars aren't like your rockstars," goes the byline of a famous Intel ad featuring Ajay Bhatt shown as having a rockstar status at the company. Not only at Intel, Ajay is considered as tech's rockstar for co-inventing the USB which has made our lives so much easier and less complicated. He believed that there should be a universal port for all peripheral devices when he was trying to connect a multimedia card to his computer. He headed the team at Intel that worked on the standards of the Universal Serial Bus (USB) which was started by a group of seven companies in 1994. The impact has been huge and today there are about 6 billion USB devices of different types around the world. He also developed the Accelerated Graphics Port (AGP) used to connect a video card to the motherboard, replacing existing PCI connections. He then went on to develop the PCIe bus standard (short for PCI Express), a high-speed bus standard that replaced all the previous PCI, PCI-X and AGP bus standards.

3 Edward Roberts (Personal Computer)



After the Altair 8800, the world's first personal computer, was featured on the January 1975 cover of *Popular Electronics* magazine, a revolution started off in the field of computing. Earlier, computers were only meant for academic and military purposes and not for personal use. Roberts, however, wanted computers to be something more and so with his experience in electronics, he decided to build a personal computer. The Altair, which needed to be assembled, was mostly purchased by hobbyists. It was a great success and the company gained huge profits. It also saw the origin of Microsoft when this machine caught the eyes of Bill Gates and Paul Allen. The duo persuaded Roberts to develop software for the Altair using a new computer language called BASIC. What followed was a rivalry between Apple and IBM, and many more companies later, to build the best commercial personal computer in the market.

4 Jonathan Fletcher (Search Engine)



On graduating from the University of Stirling, Scotland, Jonathan worked as a systems administrator at the university itself and created the world's first web-crawling search engine, JumpStation way back in '93. Though internet search engines were already operative for a couple of years, JumpStation was the first to search the World Wide Web. This search engine was based on the same